BITS Pilani

Deep Neural Network
Module # 9: Transformer

Seetha Parameswaran
WILPD - CSIS

Work Integrated Learning Programmes

The instructor is gratefully acknowledging
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This deck is prepared by Seetha Parameswaran.

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Recap: Attention Mechanisms

- What We Learned:
  - Attention allows models to focus on relevant parts of input
  - Query-Key-Value framework: Attention as information retrieval
  - Multiple attention types:
    - Self-attention: Q, K, V from same sequence
    - Multi-head: Parallel attention with different projections
    - Cross-attention: Q from one sequence, K,V from another
    - Masked: Prevent attending to future tokens
- What's Missing?
  - How to organize these attention mechanisms into a complete architecture?
  - How to handle sequences without recurrence?
  - How to scale to billions of parameters?
- This Module:
  - Transformer architecture: Attention as the only mechanism
  - Three variants: Encoder-only, Decoder-only, Encoder-decoder

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THE RNN BOTTLENECK

- Recurrent Neural Networks (RNNs):
  - Process sequences one token at a time: $h_t = f(h_{t-1}, x_t)$
  - Hidden state carries information forward
  - Used with attention
- Three Critical Problems:
  - **Sequential Processing**: Cannot parallelize
    - Must compute h_1 before h_2 before h_3...
    - Training is slow (hours $\rightarrow$ days $\rightarrow$ weeks)
  - **Vanishing Gradients**: Long-range dependencies fail
    - Information from early tokens fades
    - Even with LSTM/GRU, struggles beyond 100-200 tokens
  - **Fixed Memory**: Hidden state is bottleneck
    - All information compressed into fixed-size vector
    - Limited capacity as sequences grow
- Question: Can we build sequence models without recurrence?

THE TRANSFORMER SOLUTION (2017)

- Key Insight: "Attention is All You Need"
  - Replace recurrence entirely with self-attention
  - Each token directly attends to all other tokens
  - No sequential dependency → full parallelization
- How Transformers Solve RNN Problems:

Problem	RNN	Transformer
Parallelization	Sequential	Parallel
Long-range deps	Vanishing gradients	Direct connections
Memory	Fixed hidden state	Attention to all tokens

- The Impact:
  - 2017: Machine translation breakthrough
  - 2018: BERT revolutionizes NLP (encoder-only)
  - 2018-2020: GPT series (decoder-only)
  - 2020-present: Foundation of all large language models
- Result: Transformers became universal architecture for sequences

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From Transformers to Modern AI

- Transformers Enable Scaling:
  - Parallel processing → train on massive datasets
  - Direct connections → learn complex patterns
  - Flexible architecture → adapt to any domain
- This Scaling Led To:
  1. **Large Language Models (LLMs)**
     - 100B+ parameters: GPT-4, Claude, Gemini
     - Trained on trillions of tokens
     - General intelligence: reasoning, coding, writing
  2. **Small Language Models (SLMs)**
     - 1B-7B parameters: Phi-3, Gemma, Mistral-7B
     - On-device deployment: phones, laptops
     - Privacy-preserving, low-latency
  3. **Specialized Models**
     - Vision: ViT, CLIP (images without convolutions)
     - Speech: Whisper, Wav2Vec2 (audio understanding)
     - Multimodal: GPT-4V, Gemini (text + images + audio)

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Real-world Applications

1. Conversational AI
   - ChatGPT, Claude, Gemini: Natural dialogue with context
   - Customer support automation: 24/7 intelligent responses
   - Virtual assistants: Multi-turn conversations with memory
2. Code Generation & Software Development
   - GitHub Copilot: Real-time code completion
   - CodeLlama, StarCoder: Function generation from descriptions
3. Content Creation
   - Writing assistance: Articles, emails, reports
   - Summarization: Long documents $\rightarrow$ key points
   - Translation: 100+ languages with context
4. Scientific Research
   - AlphaFold 3: Protein structure prediction
   - Drug discovery: Molecular design and screening
5. Next: How do these models differ? Three transformer variants

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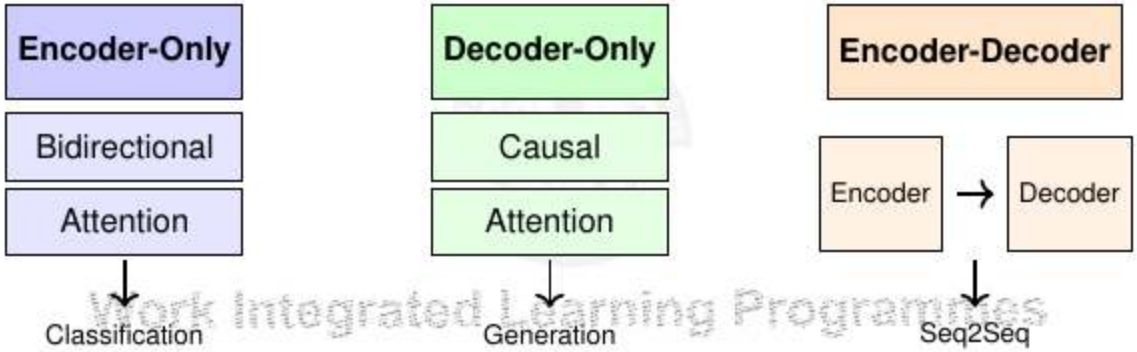
Three Transformer Architectures

- Why Three Variants?
  - Different tasks need different information flow
  - Trade-off between understanding vs generation
  - Computational efficiency for specific use cases



- Selection Criteria:
  - Need to classify or extract? → Encoder-only
  - Need to generate text? → Decoder-only
  - Input and output very different? → Encoder-decoder

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Industry Models Comparison

Model	Arch	Parameters	Strengths	Access
GPT-4	Decoder	1.8T	Reasoning, code	API (OpenAI)
Claude 4	Decoder	200B	Safety, long context	API (Anthropic)
Gemini 3	Decoder	175B	Multimodal	API (Google)
LLaMA 3	Decoder	8B-405B	Open, efficient	Open (Meta)
Mistral	Decoder	7B-123B	Fast inference	Open (Mistral AI)
BERT	Encoder	110M-340M	Understanding, NER	Open (Google)
T5	Enc-Dec	60M-11B	Translation, summary	Open (Google)

- Key Observations:
  - Decoder-only dominates generative tasks
  - Open models (LLaMA, Mistral) competitive with proprietary
  - Parameter count vs performance: Not always linear

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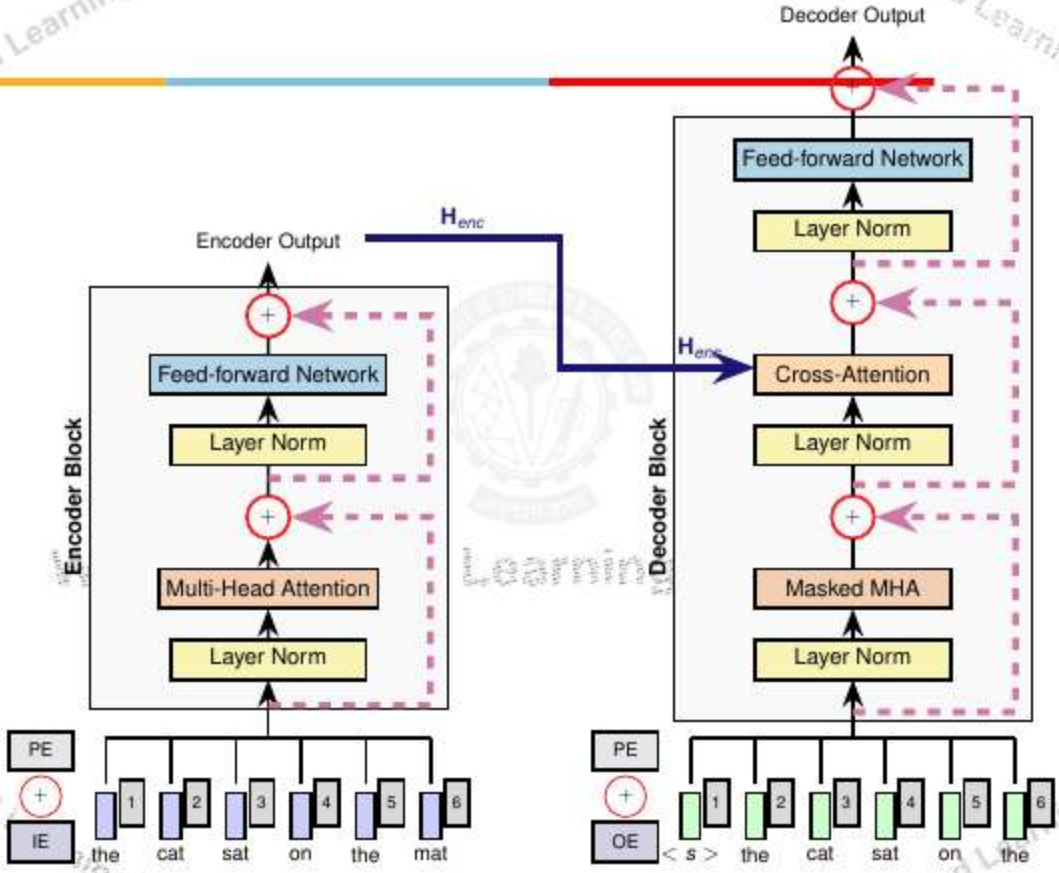
TRANSFORMER ARCHITECTURE



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What Does the Encoder Do?

- Purpose: Transform input sequence into rich contextual representations
  - Input: Raw tokens (words, subwords, patches)
  - Output: Context-aware embeddings for each input position
- Key Capability: Bidirectional Context
  - Each token attends to **all other tokens** (no restrictions)
  - Captures relationships in both directions
  - Example: "bank" in "river bank" vs "bank account"
- How it Works:
  - Multi-Head Attention: Tokens "communicate" with each other
  - Feed-Forward Network: Each token "thinks" independently
  - Stacked N times: Builds increasingly abstract representations
- Output:
  - $\mathbf{H}_{enc} \in \mathbb{R}^{n \times d_{\text{model}}}$ where n is sequence length
  - Each row is a contextualized representation of one input token
  - Used by decoder (encoder-decoder) or task head (encoder-only)

What Does the Decoder Do?

- Purpose: Generate output sequence one token at a time
  - Input: Previously generated tokens (autoregressive)
  - Output: Probability distribution over next token
- Key Capability: Causal Generation
  - Token at position t can only attend to positions $\leq t$
  - Prevents "looking ahead" to future tokens
  - Ensures model learns to predict sequentially
- How it Works:
  - Masked Self-Attention: Tokens attend only to past
  - Cross-Attention: Attends to encoder output (encoder-decoder only)
  - Feed-Forward Network: Process information independently
  - Stacked N times: Refines generation at each layer
- Output:
  - $\mathbf{D}_{out} \in \mathbb{R}^{m \times d_{\text{model}}}$ where m is target length
  - Select next token (greedy/sampling/beam search)

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Transformer Encoder: Input Embedding

- Input Embedding:
  - Convert tokens to dense vectors: $\mathbf{x}_i \in \mathbb{R}^{d_{\text{model}}}$
  - Learned embedding matrix: $\mathbf{E} \in \mathbb{R}^{|V| \times d_{\text{model}}}$
  - Typical $d_{\text{model}} = 512$ (base) or 1024 (large)

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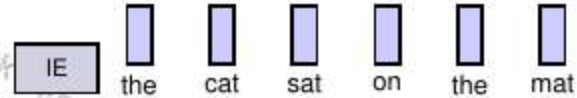
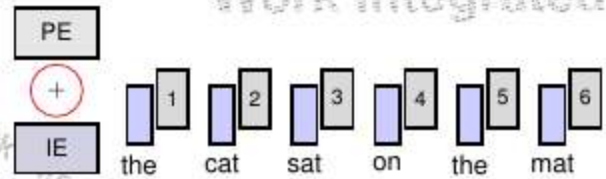


Transformer Encoder: Position Embedding

- Positional Encoding:
  - Problem: Attention has no notion of position
  - Solution: Add position information to embeddings

$$\mathrm{PE}(\mathrm{pos}, 2i) = \sin\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right) \tag{1}$$

$$\mathrm{PE}(\mathrm{pos}, 2i+1) = \cos\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\right) \tag{2}$$



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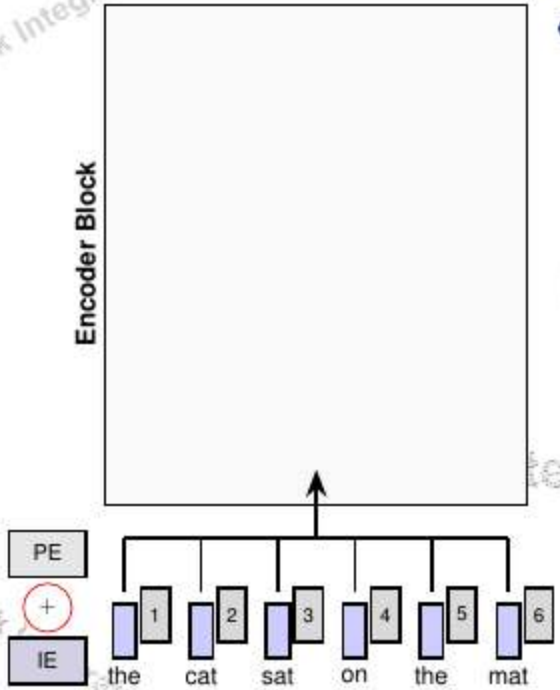
Transformer Encoder: Encoder Input



- Final Encoder Input: = Input Embedding concatenate Position Embedding

$$\mathbf{X} = \text{Embedding}(\text{tokens}) + \text{PE}(\text{positions}) \tag{3}$$

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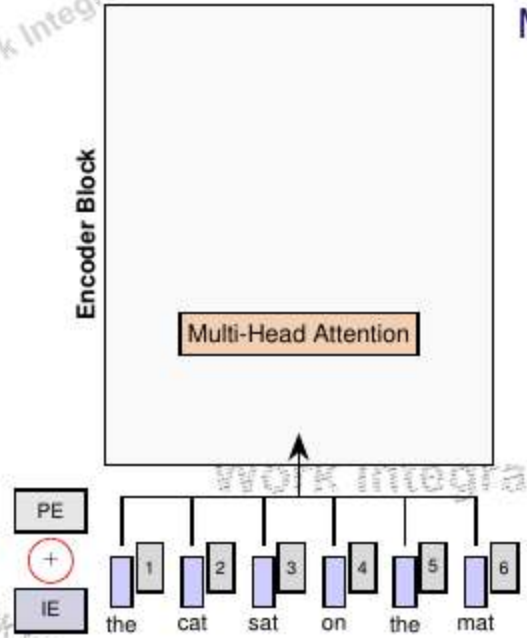
TRANSFORMER ENCODER: MHA



Multi-Head Attention:

- Allows each embedding to attend to all other embeddings
- h parallel attention heads capture different relationships
- **For each head** $h = 1, \ldots, H$:

$$\mathbf{Q}_h = \mathbf{Z}\mathbf{W}_h^Q, \quad \mathbf{K}_h = \mathbf{Z}\mathbf{W}_h^K, \quad \mathbf{V}_h = \mathbf{Z}\mathbf{W}_h^V \tag{4}$$

$$\text{head}_h = \text{softmax}\left(\frac{\mathbf{Q}_h\mathbf{K}_h^T}{\sqrt{d_k}}\right)\mathbf{V}_h \tag{5}$$

$$\text{MHA}(\mathbf{Z}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_H)\mathbf{W}^O \tag{6}$$

- **Typical:** $H = 12$ heads, $d_k = d_v = D/H = 64$

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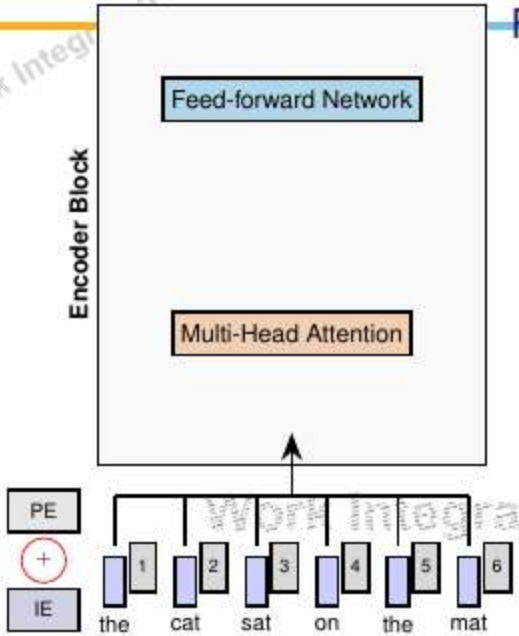
TRANSFORMER ENCODER: FFN



Feed-Forward Network (Position-wise):

- FFN "Thinks" **independently** and processes the information
- Two-layer FFN with expansion
  - **Expand:** $D \to 4D$; $\mathbf{W}_1 \in \mathbb{R}^{D \times 4D}$ parameters
  - **Non-linearity:** RELU activation
  - Applied **independently** to each position
  - **Contract:** $4D \to D$; $\mathbf{W}_2 \in \mathbb{R}^{4D \times D}$ parameters
- Why Position-wise FFN?
  - Same weights applied at each position independently
  - Adds non-linearity after attention
  - Provides model capacity for complex transformations

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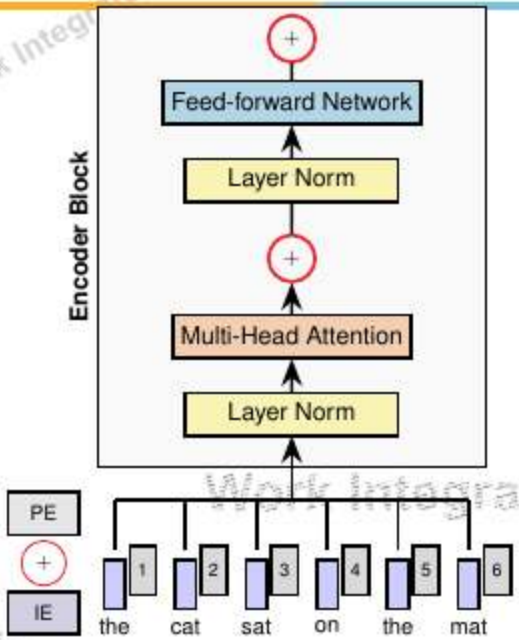
Transformer Encoder: Layer Norm



- Problem: Training Deep Networks is Hard
  - Stacking many layers → vanishing gradients
  - Information loss
  - Optimization becomes unstable
- Solution 1: Layer Normalization
  - Normalize across feature dimension
  - Stabilizes training across features (not batch)
  - Learnable parameters: $\gamma, \beta \in \mathbb{R}^{d_{\text{model}}}$
  - Same procedure at train/test time

$$\text{LN}(\mathbf{x}) = \gamma \frac{\mathbf{x} - \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}} + \beta \tag{7}$$

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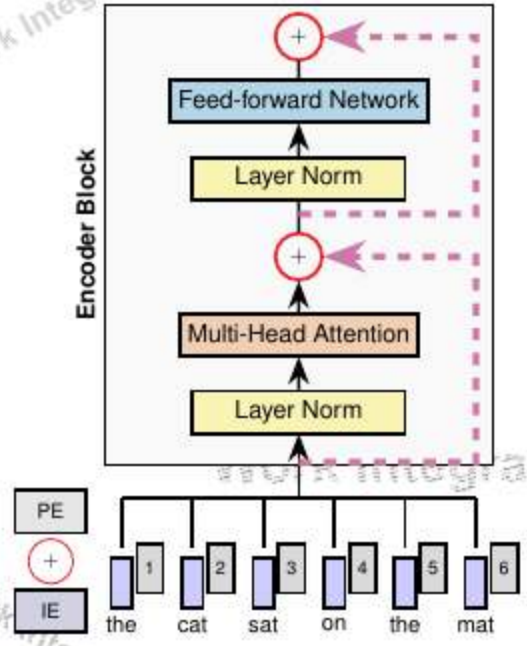
Transformer Encoder: Residual Connections



- Problem: Training Deep Networks is Hard
- Solution 2: Residual Connections (Skip Connections)
  - Help gradient flow in deep networks (from ResNet)
  - **Add input directly to output:**

$$\text{Out} = \mathbf{x} + \text{Sublayer}(\mathbf{x}) \tag{8}$$

  - Creates "gradient highways" → easier optimization
  - Allows training very deep models (100+ layers)

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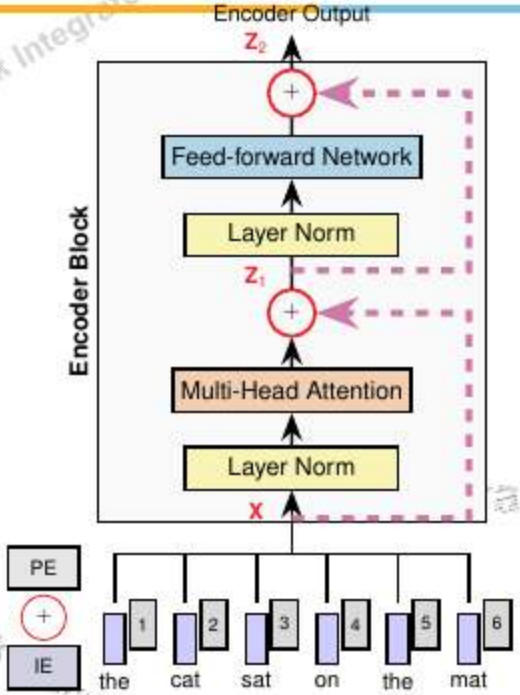
Transformer Encoder: Encoder Output



- Complete Encoder Layer:

$$\mathbf{Z}_1 = \mathbf{X} + \text{MHA}(\text{LayerNorm}(\mathbf{X})) \quad (9)$$

$$\mathbf{Z}_2 = \mathbf{Z}_1 + \text{FFN}(\text{LayerNorm}(\mathbf{Z}_1)) \quad (10)$$

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Transformer Encoder: Parameters

Each token gets embedded into a d_{model}-dimensional vector, and we can process sequences up to $n_{\max}$ tokens long.

- $|V|$: Vocabulary size (30K-50K tokens)
- d_{model}: Embedding vector length/dimension (512 (base) or 1024 (large))
- $n_{\max}$: Maximum sequence length (512-2048 tokens)

Input Embedding:	$\lvert V\rvert \times d_{\text{model}}$
Positional Encoding:	$n_{\max} \times d_{\text{model}}$ (learnable)
Layer Norm (2 layers):	$2 \times 2 \times d_{\text{model}} = 4d_{\text{model}}$
Multi-Head Attention:	$4 \times d_{\text{model}}^2$ (for W^Q, W^K, W^V, W^O)
Feed-Forward Network:	$2 \times d_{\text{model}} \times d_{\text{ff}} = 8d_{\text{model}}^2$ where $d_{\text{ff}} = 4d_{\text{model}}$
	$W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}, \quad W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$
Total per layer:	$4d_{\text{model}} + 4d_{\text{model}}^2 + 8d_{\text{model}}^2 \approx 12d_{\text{model}}^2$

ENCODER STACK

- Stack of N Identical Layers:
  - Original Transformer: $N = 6$ layers
  - BERT-base: $N = 12$ layers
  - BERT-large: $N = 24$ layers
  - GPT-3: $N = 96$ layers (decoder, but same principle)
- Stacking Process:

$$\mathbf{H}^{(\ell)} = \text{EncoderLayer}^{(\ell)}(\mathbf{H}^{(\ell-1)}), \quad \ell = 1, \ldots, N \tag{11}$$

where $\mathbf{H}^{(0)} = \text{InputEmbedding} + \text{PositionalEncoding}$

- Encoder Output:
  - Final encoder output: $\mathbf{H}^{(N)} \in \mathbb{R}^{n \times d_{\text{model}}}$
  - Rich contextual representation for each input token
  - Each position has attended to all other positions N times
  - Passed to decoder via cross-attention (in encoder-decoder models)

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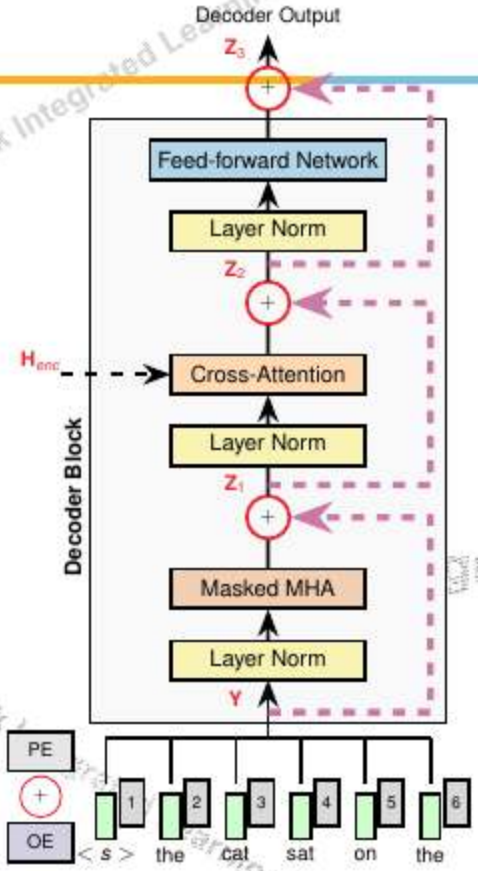
Transformer Decoder: Decoder Output



- Complete Decoder Layer:

$$\mathbf{Z}_1 = \mathbf{Y} + \text{MaskedMHA}(\text{LayerNorm}(\mathbf{Y})) \quad (12)$$

$$\mathbf{Z}_2 = \mathbf{Z}_1 + \text{CrossAttn}(\text{LayerNorm}(\mathbf{Z}_1), \mathbf{H}_{enc}) \quad (13)$$

$$\mathbf{Z}_3 = \mathbf{Z}_2 + \text{FFN}(\text{LayerNorm}(\mathbf{Z}_2)) \quad (14)$$

where $\mathbf{H}_{enc}$ is encoder output

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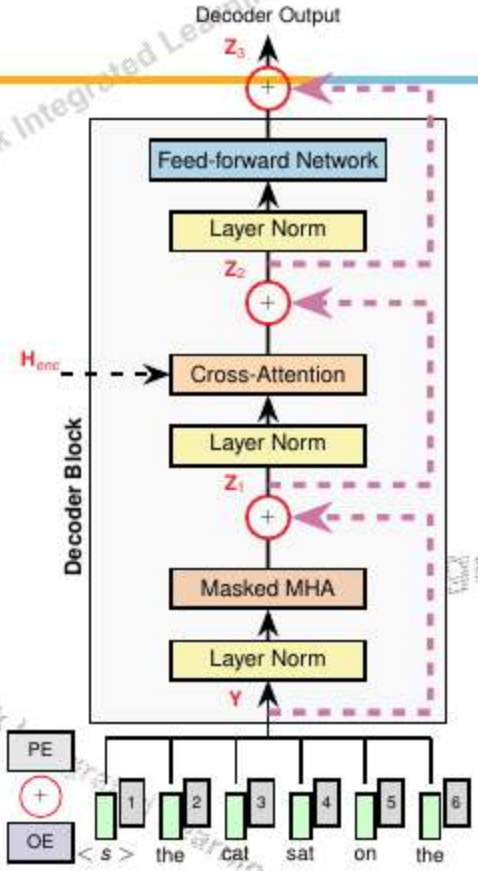
Transformer Decoder: Masked Multi-Head Attention



- Self-attention with causal masking
  - Token t attends only to positions $\leq t$
  - Prevents "looking ahead" to future tokens
- **Why?** Match training and inference
  - At inference: Generate one token at a time
  - Training must simulate this autoregressive process

$$\text{mask}_{ij} = \begin{cases} 0 & \text{if } i \geq j \\ -\infty & \text{if } i < j \end{cases} \tag{15}$$

$$\text{MaskedAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \text{mask}\right)\mathbf{V} \tag{16}$$

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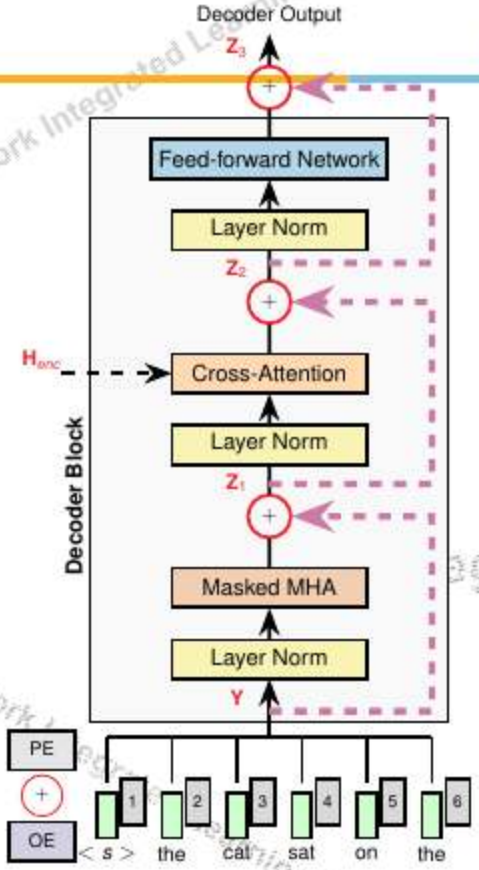
Transformer Decoder: Cross-Attention



Cross-Attention (Encoder-Decoder Attention):

- Decoder attends to encoder output
  - Query (Q): From decoder layer
  - Key (K), Value (V): From encoder output $\mathbf{H}_{enc}$
  - No masking (can attend to all source tokens)
- **Why needed:** Connect source and target
  - Allows decoder to focus on relevant source positions

$$\text{CrossAttn}(\mathbf{Q}_{dec}, \mathbf{K}_{enc}, \mathbf{V}_{enc}) = \text{softmax}\left(\frac{\mathbf{Q}_{dec}\mathbf{K}_{enc}^T}{\sqrt{d_k}}\right)\mathbf{V}_{enc}$$

$$\mathbf{Q}_{dec} = \text{LN}(\mathbf{Z}_1)\mathbf{W}^Q$$

$$\mathbf{K}_{enc} = \mathbf{H}_{enc}\mathbf{W}^K$$

$$\mathbf{V}_{enc} = \mathbf{H}_{enc}\mathbf{W}^V$$

Transformer Decoder: Parameters

Each output token gets embedded into a d_{model}-dimensional vector.

- Same vocabulary and embedding dimensions as encoder
- $|V|$, d_{model}, $n_{\max}$ same as encoder

Output Embedding:	$\lvert V\rvert \times d_{\text{model}}$
Positional Encoding:	$n_{\max} \times d_{\text{model}}$ (learnable)
Layer Norm (3 layers):	$3 \times 2 \times d_{\text{model}} = 6d_{\text{model}}$
Masked Multi-Head Attention:	$4 \times d_{\text{model}}^2$ (for W^Q, W^K, W^V, W^O)
Cross-Attention:	$4 \times d_{\text{model}}^2$ (for W^Q, W^K, W^V, W^O)
Feed-Forward Network:	$2 \times d_{\text{model}} \times d_{\text{ff}} = 8d_{\text{model}}^2$ where $d_{\text{ff}} = 4d_{\text{model}}$
	$W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}, \quad W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$
Total per layer:	$6d_{\text{model}} + 8d_{\text{model}}^2 + 8d_{\text{model}}^2 \approx 16d_{\text{model}}^2$

Note: Decoder has 33% more parameters than encoder (16 vs 12 d_{model}^2)

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Decoder Stack

- Stack of N Identical Decoder Layers:
  - Original Transformer: $N = 6$ layers
  - Each layer has **3 sublayers** (vs 2 in encoder):
    1. Masked multi-head self-attention
    2. Cross-attention to encoder output
    3. Position-wise feed-forward network
- Stacking Process:

$$\mathbf{D}^{(\ell)} = \text{DecoderLayer}^{(\ell)}(\mathbf{D}^{(\ell-1)}, \mathbf{H}_{enc}), \quad \ell = 1, \ldots, N \tag{17}$$

where $\mathbf{D}^{(0)} = \text{OutputEmbedding} + \text{PositionalEncoding}$

- Decoder Output:
  - Final decoder output: $\mathbf{D}^{(N)} \in \mathbb{R}^{m \times d_{\text{model}}}$ where m is target length
  - Each position has attended to previous positions N times (masked)
  - Each position has attended to all encoder positions N times (cross-attention)

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Output Layer: Linear + Softmax

- Purpose: Convert decoder output to vocabulary probabilities
- Linear Projection:

$$\text{logits} = \mathbf{D}^{(N)} W_{\text{out}} + b_{\text{out}} \tag{18}$$

where $W_{\text{out}} \in \mathbb{R}^{d_{\text{model}} \times |V|}$, $b_{\text{out}} \in \mathbb{R}^{|V|}$

- Softmax: Output: Probability distribution over vocabulary

$$P(y_t | y_{<t}, \mathbf{x}) = \text{softmax}(\text{logits}_t) \tag{19}$$

- Complete Flow:

$$\mathbb{R}^{d_{\text{model}}} \xrightarrow{\text{Linear}} \mathbb{R}^{|V|} \xrightarrow{\text{Softmax}} \text{Probabilities} \xrightarrow{\text{Token Selection}} y_t$$

- Token selection done via greedy, sampling, or beam search

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From NLP to Vision - Why ViTs Work

- Traditional Approach: CNNs
  - Convolutional layers extract local features
  - Pooling layers for spatial hierarchy
  - Inductive bias: Translation equivariance, locality
- The ViT (Vision Transformer) Innovation (2020):
  - Treat image as sequence of patches
  - Apply pure transformer (no convolutions)
  - Learn visual relationships through self-attention
- Key Insight:
  - CNNs: Hard-coded locality through convolution kernel
  - ViTs: Learn relationships between all patches
  - Trade-off: Need more data, but more flexible
- When ViTs Excel:
  - Large-scale pretraining (ImageNet-21K, JFT-300M)
  - Transfer learning to downstream tasks

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What Does Vision Transformer (ViT) Do?

- Purpose: Image classification using pure transformer (no convolutions)
- Architecture: Encoder-only (no decoder needed for classification)
- Input → Output:
- Key Innovation: Treat image as sequence of patches
  - Divide image into fixed-size patches (16×16)
  - Flatten each patch → Linear projection
  - Add positional encoding
  - Process through transformer encoder
- How Classification Works:
  - Prepend [CLS] token to patch sequence
  - Transformer encoder processes all patches
  - .[CLS] output → MLP head → Class probabilities
- Why It Works: Self-attention captures spatial relationships globally

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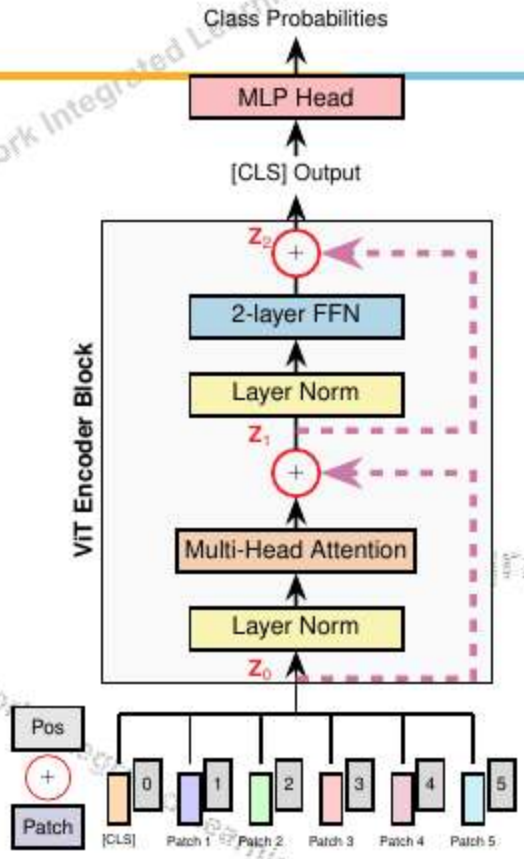
Vision Transformer (ViT): Architecture



- Architecture: ~~Encoder-only (no~~ decoder)
  - Input: Image → Output: Class probabilities
  - Uses only transformer encoder blocks
  - No decoder needed for classification task
- Classification [CLS] Token:
  - Learnable special token prepended to patch sequence
  - Aggregates global image information via self-attention
  - Final [CLS] representation used for classification
- Processing Flow:

  Image → Patches → Encoder → [CLS] → Class

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Patch Embedding & [CLS] Token

- Image to Patches:
  - Input image: $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ (e.g., $224 \times 224 \times 3$)
  - Divide into N patches of size $P \times P$ (typically $P = 16$)
  - Number of patches: $N = \frac{H \times W}{P^2} = \frac{224 \times 224}{16 \times 16} = 196$
- Linear Projection:

$$\mathbf{z}_i = \text{Flatten}(\text{patch}_i) W_p, \quad W_p \in \mathbb{R}^{(P^2 \cdot C) \times d} \tag{20}$$

  - Flatten each patch: $16 \times 16 \times 3 = 768$ pixels
  - Project to embedding dimension d (typically 768)
  - Result: $\mathbf{Z} = [\mathbf{z}_1; \mathbf{z}_2; \ldots; \mathbf{z}_N] \in \mathbb{R}^{N \times d}$
- Prepend [CLS] Token:
  - Learnable embedding $\mathbf{z}_{[\text{CLS}]} \in \mathbb{R}^d$ prepended to sequence
  - Aggregates global image information via self-attention
  - Final sequence: $[\mathbf{z}_{[\text{CLS}]}; \mathbf{z}_1; \mathbf{z}_2; \ldots; \mathbf{z}_N] \in \mathbb{R}^{(N+1) \times d}$

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Positional Encoding

- Problem: Self-attention is permutation-invariant
  - Model cannot distinguish patch positions without explicit encoding
  - Order of patches matters for understanding image structure
- Solution: Learnable Positional Embeddings
  - $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1)\times d}$ where $N + 1 = 197$ positions
  - One embedding for each position: [CLS] (position 0) + 196 patches
  - Learned during training (not fixed like sinusoidal in transformers)
  -
  - 1D embeddings sufficient; no 2D structure hardcoded
- Adding Position Information:

$$\mathbf{z}_0^{(0)} = [\mathbf{z}_{[\text{CLS}]}; \mathbf{z}_1; \mathbf{z}_2; \ldots; \mathbf{z}_N] + \mathbf{E}_{\text{pos}} \tag{21}$$

  - Element-wise addition of position embeddings
  - Input to first encoder layer: $\mathbf{z}_0^{(0)} \in \mathbb{R}^{(N+1)\times d}$

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ViT Encoder: Processing

- Encoder Layer Operations:
  - Same as Transformer encoder:
    Layer Norm $\rightarrow$ MHA $\rightarrow$ Add $\rightarrow$ Layer Norm $\rightarrow$ FFN $\rightarrow$ Add
  - Stack of L identical layers (ViT-Base: $L = 12$)
- Layer ℓ aligns: $\mathbf{z}_0^{(\ell-1)} \in \mathbb{R}^{(N+1)\times d}$ is input to layer ℓ

$$\mathbf{z}_1^{(\ell)} = \mathbf{z}_0^{(\ell-1)} + \text{MHA}(\text{LayerNorm}(\mathbf{z}_0^{(\ell-1)})) \tag{22}$$

$$\mathbf{z}_2^{(\ell)} = \mathbf{z}_1^{(\ell)} + \text{FFN}(\text{LayerNorm}(\mathbf{z}_1^{(\ell)})) \tag{23}$$

- Multi-Head Self-Attention:
  - Each patch (including [CLS]) attends to all other patches
  - Captures global spatial relationships
  - ViT-Base: $H = 12$ heads, $d_k = d/H = 64$
- Feed-Forward Network: $W_1 \in \mathbb{R}^{d\times 4d}$, $W_2 \in \mathbb{R}^{4d\times d}$ (expansion factor 4)

$$\text{FFN}(\mathbf{x}) = \text{GELU}(\mathbf{x}W_1 + b_1)W_2 + b_2 \tag{24}$$

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ViT Output: Classification

- Final Encoder Output:
  - After L layers: $\mathbf{z}_2^{(L)} \in \mathbb{R}^{(N+1)\times d}$
  - Extract [CLS] token representation: $\mathbf{z}_{[\text{CLS}]}^{(L)} \in \mathbb{R}^d$
  - This contains aggregated information from entire image
- Classification Head (MLP):

$$\mathbf{h} = \text{LayerNorm}(\mathbf{z}_{[\text{CLS}]}^{(L)}) \tag{25}$$

$$\text{logits} = \mathbf{h}W_{\text{head}} + b_{\text{head}} \tag{26}$$

where $W_{\text{head}} \in \mathbb{R}^{d\times K}$, K is number of classes

- Softmax for Probabilities:

$$P(y = k|\mathbf{I}) = \text{softmax}(\text{logits})_k \tag{27}$$

  - Output: Probability distribution over K classes
  - Final prediction: $\hat{y} = \arg\max_k P(y = k|\mathbf{I})$
- Training: Cross-entropy loss between predictions and true labels

ViT vs CNNs - Trade-offs

Aspect	CNNs	ViT
Inductive Bias	Strong (locality, translation equivariance)	Weak (learns from data)
Data Efficiency	Good with small datasets	Requires large-scale pre-training
Scalability	Limited by kernel size	Scales well with model size
Global Context	Limited (small receptive field)	Excellent (self-attention)
Compute	Efficient convolutions	Quadratic in sequence length
Transfer Learning	Good	Excellent (better than CNNs)

- Practical Guidelines:
  - **Use CNNs**: Small datasets (<100K images), limited compute
  - **Use ViT**: Large-scale pretraining available, transfer learning
  - **Hybrid**: Combine CNN stem with Transformer body

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Why Three Transformer Variants?

- Different Tasks Need Different Architectures
  - Not all NLP tasks require generation
  - Not all tasks need bidirectional context
  - Trade-offs between capability and efficiency
- Key Design Decision: Attention Pattern

Aspect	Encoder-Only	Decoder-Only	Encoder-Decoder
Attention	Bidirectional	Causal (masked)	Both
Information Flow	All tokens see all	Token sees only past	Encoder: bi, Decoder: causal
Task Type	Understanding	Generation	Transformation
Use Case	Full context needed	Sequential output	Input $\neq$ Output
Efficiency	Single pass, parallel	Simpler, scalable	Most flexible, compute-intensive

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Encoder-Only Architecture

- Architecture:
  - Only encoder stack (no decoder)
  - Bidirectional self-attention: Each token attends to all tokens
  - Output: Rich contextual representations for each input token
- Pretraining: Masked Language Modeling (MLM)
  - Randomly mask 15% of input tokens
  - Predict masked tokens using context from both directions
  - Example: "The [MASK] sat on the mat" → predict "cat"
- Use Cases:
  - Text classification: Sentiment analysis, spam detection
  - Named Entity Recognition (NER): Extract names, locations
  - Question Answering: Extract answer spans from context
  - Sentence embeddings: Semantic similarity, clustering

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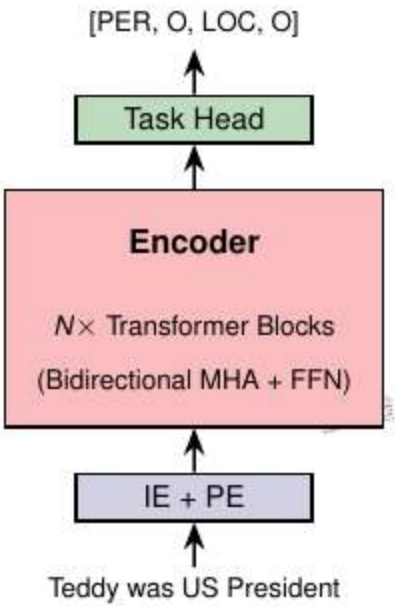
Transformer: Encoder-Only Architecture



Popular Models:

- BERT (2018): 110M-340M parameters
- RoBERTa (2019): Optimized BERT training
- DistilBERT (2019): 40% smaller, 60% faster

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Decoder-Only Architecture

- Architecture:
  - Only decoder stack (no encoder)
  - Causal/masked self-attention: Token t attends only to $\leq t$
  - Output: Probability distribution over next token
- Pretraining: Autoregressive Language Modeling
  - Predict next token given previous tokens
  - Training objective: $P(x_t|x_{<t})$
  - Example: "The cat sat on" $\rightarrow$ predict "the"
- Use Cases:
  - Text generation: Creative writing, content creation
  - Conversational AI: ChatGPT, Claude, Gemini
  - Code generation: GitHub Copilot, CodeLlama
  - Few-shot learning: In-context learning without fine-tuning

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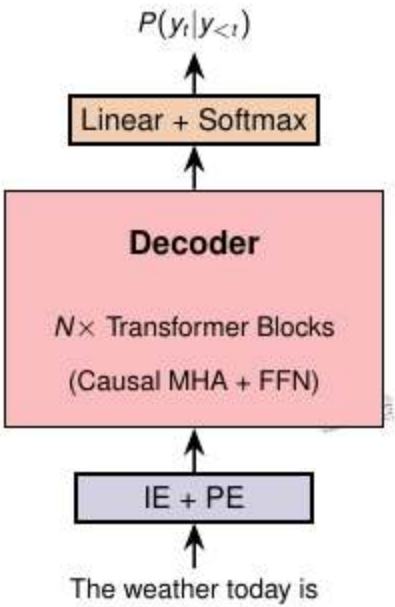
Transformer: Decoder-Only Architecture



Popular Models:

- GPT-4 (2023): 1.8T parameters (estimated)
- Claude 4 (2025): 200B parameters
- LLaMA 3 (2024): 8B-405B parameters (open)
- Mistral (2023): 7B-123B parameters (open)

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Encoder-Decoder Architecture

- Architecture:
  - Full transformer: Both encoder and decoder stacks
  - Encoder: Bidirectional self-attention on input
  - Decoder: Causal self-attention + cross-attention to encoder
- Pretraining: Various Objectives
  - T5: Text-to-text format (all tasks as generation)
  - BART: Denoising autoencoder (corrupt and reconstruct)
  - mT5: Multilingual text-to-text
- Use Cases:
  - Machine translation: English → French, etc.
  - Summarization: Long document → concise summary
  - Question answering: Question + context → answer
  - Data-to-text: Structured data → natural language

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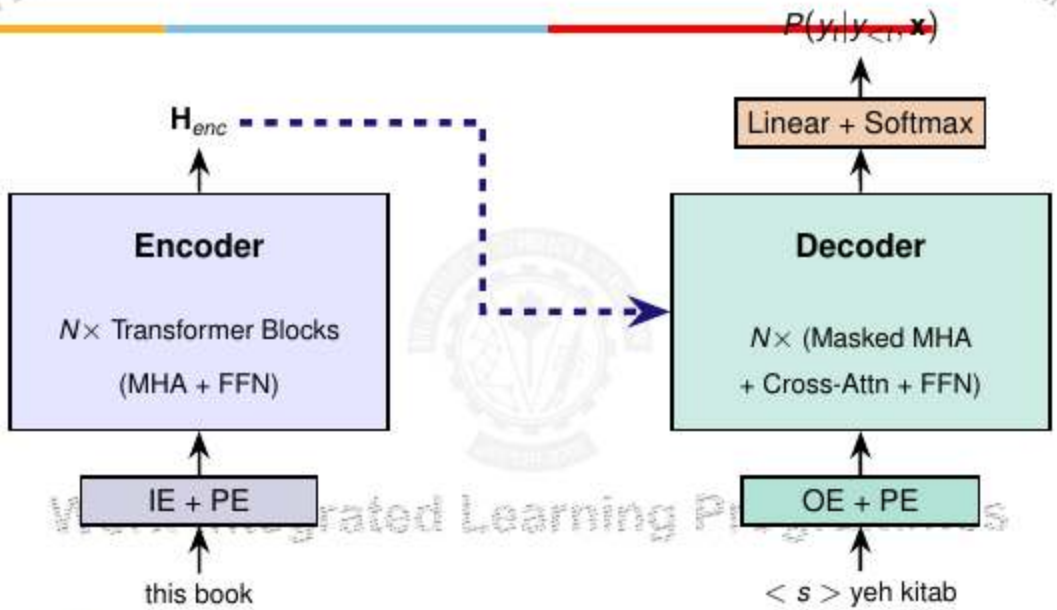
Encoder-Decoder Transformer



Popular Models:

- T5 (2019): 60M-11B parameters
- mT5 (2020): Multilingual T5 (101 languages)

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Training vs Inference: Examples

- **Encoder-Decoder (Translation):**
  - Training:
    - Input: "this book is good" → Target: "yeh kitab acchi hai"
    - Decoder sees all target tokens with masking
    - Predicts all tokens in parallel (teacher forcing)
  - Inference:
    - Step 1: ["<start>"] → "yeh"
    - Step 2: ["<start>", "yeh"] → "kitab" (continues...)
- **Encoder-Only (Named Entity Recognition):**
  - Input 1: "Teddy is a soft toy" → Output: [O, O, O, O, O] (O=other)
  - Input 2: "Teddy was US President" → Output: [PER, O, LOC, O]
  - Processes entire sentence at once (bidirectional context)
- **Decoder-Only (Text Completion):**
  - Prompt: "The weather today is"
  - Generates: "sunny and warm with a gentle breeze"
  - Each token generated based on all previous tokens

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Comparison: When to Use What

Aspect	Encoder-Only	Decoder-Only	Encoder-Decoder
Attention	Bidirectional	Causal	Both
Use Case	Understanding	Generation	Transformation
Tasks	Classification, NER, QA	Chat, completion	Translation, summary
Examples	Sentiment, entity tagging	Code gen, dialogue	Eng→Hindi, abstractive
Models	BERT, RoBERTa	GPT-4, Claude, LLaMA	T5, BART, mT5
Context	Full sentence	Left-to-right	Source: full, Target: causal
Speed	Fast (parallel)	Slow (sequential)	Medium

Selection Guide:

- Need to classify or extract? → Encoder-only
- Need to generate text? → Decoder-only
- Input language ≠ Output language? → Encoder-decoder

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Large-Scale Pretraining Paradigm

- Why Pretrain?
  - Learn general language understanding from massive unlabeled data
  - Transfer knowledge to downstream tasks with limited labeled data
  - Better performance than training from scratch
- Pretraining Objectives by Architecture:
  - **Encoder-only**: Masked Language Modeling (BERT, RoBERTa)
  - **Decoder-only**: Autoregressive LM (GPT, LLaMA)
  - **Encoder-decoder**: Span corruption, denoising (T5, BART)
- Fine-tuning Approaches:
  - Full fine-tuning: Update all parameters
  - Parameter-efficient: LoRA, adapters (update $<1\%$ params)
  - Prompting: Zero-shot, few-shot (no parameter updates)

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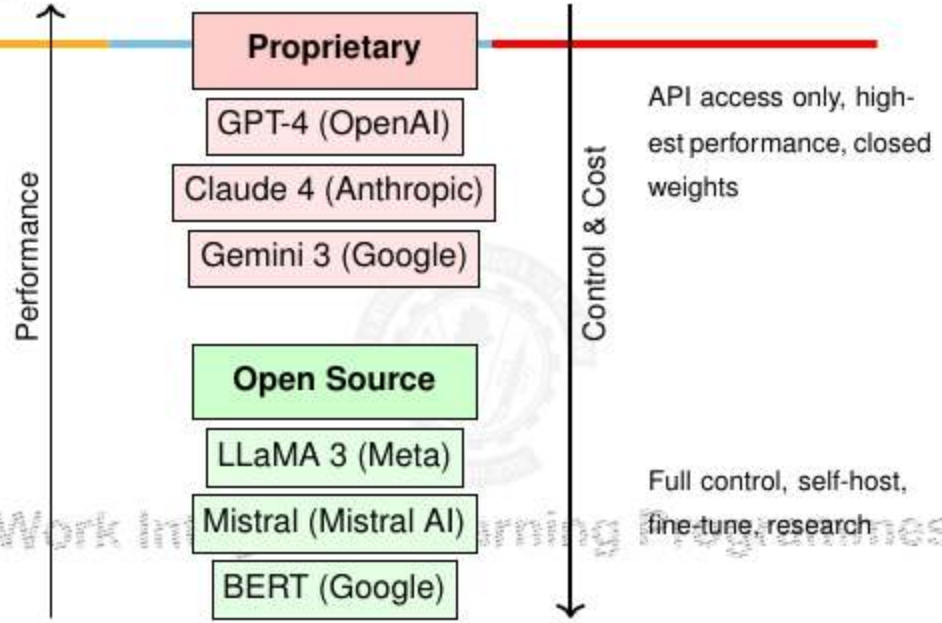
Foundation Models Landscape



- Trade-offs:
  - Proprietary: Best performance, no control, API costs
  - Open: Full control, hosting costs, requires expertise

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HuggingFace Hub

- What is HuggingFace?
  - Open platform for ML models, datasets, and applications
  - 500,000+ pre-trained models available
  - Easy-to-use `transformers` library (Python)
  - Community-driven: researchers, companies, hobbyists
- Popular Models on HuggingFace:
  - **Encoder-only**: `bert-base-uncased`, `roberta-large`, `distilbert-base`
  - **Decoder-only**: `gpt2`, `meta-llama/Llama-3-8B`, `mistralai/Mistral-7B`
  - **Encoder-decoder**: `t5-base`, `facebook/bart-large`, `google/flan-t5-large`
- Benefits: Instant access, no training needed, proven performance

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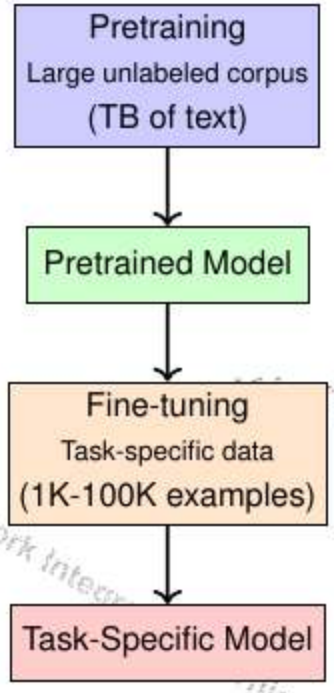
Transfer Learning & Fine-tuning

Transfer Learning Paradigm:



Self-supervised: MLM, autoregressive

Supervised: Classification, NER, etc.

Why Not Train From Scratch?

- Requires massive data (billions of tokens)
- Expensive compute (thousands of GPU-hours)
- Pretrained models already learned general language patterns

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Fine-tuning Workflow

- Step 1: Load Pretrained Model
- Step 2: Prepare Task-Specific Data
  - Tokenize inputs with pretrained tokenizer
  - Create train/validation splits
  - Prepare labels in correct format
- Step 3: Fine-tune with Small Learning Rate
  - Typical LR: 2×10^{-5} to 5×10^{-5}
  - Few epochs: 2-5 (avoid overfitting)
  - Use Adam optimizer with weight decay
- Step 4: Evaluate on Validation Set
  - Monitor metrics: Accuracy, F1, loss
  - Early stopping if validation performance degrades

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Fine-tuning Example: Sentiment Analysis

- Task: Classify movie reviews as positive/negative
- Dataset: IMDb (25K train, 25K test)
- Base Model: BERT or RoBERTa

```
from transformers import (AutoTokenizer, Trainer,
    AutoModelForSequenceClassification,  TrainingArguments)

# Load model and tokenizer
tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')
model = AutoModelForSequenceClassification.from_pretrained(
    'bert-base-uncased', num_labels=2
)

# Tokenize data
def tokenize_function(examples):
    return tokenizer(examples['text'], padding='max_length',
                     truncation=True, max_length=512)

tokenized_train = train_dataset.map(tokenize_function, batched=True
    )
tokenized_val = val_dataset.map(tokenize_function, batched=True)
```


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Fine-tuning Example: Sentiment Analysis

```
# Training arguments
training_args = TrainingArguments(
    output_dir='./results', num_train_epochs=3,
    per_device_train_batch_size=16, per_device_eval_batch_size=64,
    learning_rate=2e-5, weight_decay=0.01,
    evaluation_strategy='epoch', save_strategy='epoch',
    load_best_model_at_end=True,
)

# Trainer
trainer = Trainer(model=model, args=training_args,
    train_dataset=tokenized_train, eval_dataset=tokenized_val,
)

# Train
trainer.train()

# Evaluate
results = trainer.evaluate()
print(f"Accuracy: {results['eval_accuracy']:.4f}")
```


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Best Practices for Fine-tuning

- Learning Rate:
  - Use small learning rates: 2×10^{-5} to 5×10^{-5}
- Layer Freezing Strategies:
  - **Full fine-tuning**: Update all parameters (best performance)
  - **Gradual unfreezing**: Start with top layers, unfreeze gradually
  - **Freeze embeddings**: Only update task-specific layers
- Hyperparameter Selection:
  - Batch size: 16-32 (balance memory and convergence)
  - Epochs: 2-5 (more epochs $\rightarrow$ overfitting risk)
  - Weight decay: 0.01-0.1 (regularization)
  - Gradient clipping: Max norm 1.0 (stability)
- Data Considerations:
  - Need at least 1,000 labeled examples for fine-tuning
  - More data (10K-100K) $\rightarrow$ better performance

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Evaluation Metrics

- Classification Tasks:
  - **Accuracy**: $\frac{\text{Correct predictions}}{\text{Total predictions}}$
  - **F1 Score**: Harmonic mean of precision and recall
  - **Confusion Matrix**: Detailed breakdown of predictions
- Named Entity Recognition (NER):
  - **Entity-level F1**: Precision and recall on full entities
  - **Token-level accuracy**: Per-token classification accuracy
- Question Answering:
  - **Exact Match (EM)**: Percentage of exact answer matches
  - **F1 Score**: Token overlap between prediction and ground truth
- Text Generation:
  - **BLEU**: Precision-based metric for translation/generation
  - **ROUGE**: Recall-based metric for summarization
  - **Perplexity**: How well model predicts text

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Parameter-Efficient Fine-tuning

- Problem: Full fine-tuning updates all parameters
  - BERT-base: 110M parameters
  - Memory and compute intensive
- LoRA (Low-Rank Adaptation):
  - Freeze pretrained weights
  - Add trainable low-rank matrices: $\Delta W = BA$
  - $B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times d}$ where $r \ll d$
  - Update only $< 1\%$ of parameters

$$W_{\text{new}} = W_{\text{frozen}} + \alpha \cdot BA \tag{28}$$

- Benefits:
  - 10-100× fewer trainable parameters
  - Lower memory footprint
  - Faster training
  - Comparable or better performance

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SUMMARY

- What We Learned:
  - Transformers revolutionized AI through pure attention
  - Three architectures: Encoder-only, Decoder-only, Encoder-decoder
  - Foundation of all modern LLMs and multimodal models
  - HuggingFace makes pretrained models easily accessible
  - Fine-tuning adapts models to specific tasks efficiently
- Emerging Trends:
  - Multimodal Transformers: Text + Image + Audio (GPT-4V, Gemini)
  - Efficiency Improvements: Sparse attention, linear attention, LoRA
  - Long Context: 100K-1M token context windows (Gemini, Claude)
  - Mixture of Experts (MoE): Sparse activation for efficiency
- Practical Takeaway:
  - Start with pretrained models from HuggingFace
  - Fine-tune on your task-specific data
  - Use parameter-efficient methods for large models

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Thank you :)
All the best :)

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